

TONY WANG

U.S. CITIZEN | 425-221-4924 | wangtzero@gmail.com | [linkedin.com/in/tony-wang-66667b242](https://www.linkedin.com/in/tony-wang-66667b242)

EDUCATION

Georgia Institute of Technology
Bachelor of Science in Computer Science
Relevant Coursework: Data Structures & Algorithms, Machine Learning, Computer Organization and Programming, Discrete Math, Linear Algebra, Object-Oriented Programming (Java)

Expected May 2027
Atlanta, GA

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, SQL, Bash
AI/Backend: PyTorch, scikit-learn, vLLM, llama.cpp, FastAPI, Redis, REST APIs, SSE
Infrastructure: Linux, Kubernetes, Docker, Git, CI/CD, Prometheus, Grafana, GCP
Systems/Performance: CUDA, NUMA, cgroup v2, NCCL, NVLink, InfiniBand, PCIe, Intel PCM/MLC, AMD uProf, perf

EXPERIENCE

Cisco
Software Engineer Intern

- Profiled Cisco UCS AI systems across CPU, memory, NVMe, and GPU paths to isolate performance bottlenecks
- Engineered resumable Python/Bash benchmark harnesses for batching, KV caching, concurrency, telemetry, and reporting
- Improved dual-socket Intel CPU inference by up to **33%** using PCM/MLC to isolate NUMA and thread bottlenecks
- Orchestrated vLLM/AIPerf on Kubernetes to characterize multi-GPU scaling across latency, throughput, and utilization
- Deployed Prometheus/Grafana dashboards for real-time CPU, memory, GPU, power, and thermal telemetry

The Hong Kong University of Science and Technology
Machine Learning Research Intern

- Collaborated with physicians at Sun Yat-sen University’s First Affiliated Hospital to curate **1,000+** patient records
- Tuned Random Forest models with cross-validation, reducing overfitting and reaching a **0.79** validation R² score
- Parallelized model evaluation across **256** drug combinations, accelerating candidate ranking for clinical research
- Engineered Python/SQL/Bash pipelines preserving **99%** data integrity while reducing clinical ingestion latency by **30%**

Lenovo
Systems Software Engineer Intern

- Partnered with NVIDIA CUDA engineers to diagnose NCCL/InfiniBand latency, reducing GPU workload runtime by **20%**
- Analyzed NVLink/PCIe throughput with HPC profiling, guiding system upgrades that removed hardware bottlenecks
- Resolved recurring distributed workload failures by debugging Linux routing and DNS issues across GPU cluster nodes
- Automated Docker CI/CD validation with Python, reducing manual testing time by **40%** across validation workflows

PROJECTS

InferProf: LLM Inference Profiler | *Python, C++17, FastAPI, Docker, Linux*

- Created Linux-first LLM benchmark framework with **4** adapters, async concurrency, retries, warmups, and SSE parsing
- Implemented C++17 topology and CPU-affinity helper with sysfs and sched.setaffinity for controlled Linux runs
- Instrumented cgroup v2 CPU/memory and NVIDIA/AMD GPU utilization, VRAM, power, and energy telemetry
- Persisted atomic, resumable campaigns with deterministic IDs and reports for TTFT, tail latency, and throughput
- Established CI regression gates with **95%** bootstrap intervals, **26** automated tests, and **84%** branch-aware coverage

The Dining Council | *FastAPI, Redis, Docker, Streamlit, Gemini 2.5, Google Maps API*

- Architected multi-user FastAPI/Redis service with **10+** endpoints, atomic transactions, and TTL-based session cleanup
- Integrated Gemini/Google Maps APIs to rank **3** restaurants from **50** candidates under budget and dietary constraints
- Containerized Streamlit, FastAPI, and Redis with Docker Compose, validating **10** concurrent multi-user lobbies

Personal Website | *React, Vite, Vercel, Codex*

- Designed responsive React/Vite portfolio with custom CSS, interactive timelines, and technical project narratives
- Directed Codex agent workflows for UI implementation, code review, responsive testing, and Vercel deployment

Demo | GitHub

GitHub

Live | GitHub